REMnux Setup

Setting Up REMnux on Oracle VirtualBox

REMnux is a Linux distribution built specifically for malware analysis. This guide walks you through downloading and setting it up on Oracle VirtualBox.

What You Need

- Oracle VirtualBox installed on your host machine
- At least **4GB of RAM** to allocate to the VM (8GB recommended)
- A stable internet connection for the download

Step 1 — Download REMnux

REMnux is distributed as an OVA file (a pre-built virtual machine image) which makes setup straightforward.

1. Go to the official REMnux website: REMnux
2. Click **Get REMnux** and download the OVA file
3. Verify the download using the SHA256 hash provided on the site before importing

> Always verify the hash of security tools you download. If the hash doesn't match, discard the file and re-download.

Step 2 — Import the OVA into VirtualBox

1. Open VirtualBox
2. Go to **File → Import Appliance**
3. Click the folder icon and select the downloaded `.ova` file
4. Click **Next**
5. Review the appliance settings — you can adjust RAM and CPU here if needed
6. Click **Import** and wait for the process to finish (may take a few minutes)

Step 3 — Configure the VM before first boot

Before starting the VM, tweak a few settings:

1. Select the REMnux VM in the VirtualBox sidebar
2. Click **Settings**

System → Motherboard:

- Set base memory to at least **4096 MB** (4GB)

System → Processor:

- Set to at least **2 CPUs**

Network:

- Adapter 1 should be set to **NAT** by default — leave it as NAT for internet access during analysis
- If you want to isolate the VM from your network during detonation, you can switch this to **Host-only Adapter** — but NAT is fine to start with since we manually bring the interface down before detonating

3. Click **OK** to save

Step 4 — Start the VM and log in

1. Double-click the REMnux VM to start it
2. Default credentials:
   - **Username:** `remnux`
   - **Password:** `malware`

Step 5 — Update the system

Once logged in, open a terminal and run a full system update first:

```
sudo apt update && sudo apt upgrade -y
```

This may take a few minutes depending on your connection. Let it finish completely before moving on.

Step 6 — Install inotify-tools

REMnux does not include `inotify-tools` by default. Install it now:

```
sudo apt install inotify-tools -y
```

Verify it installed correctly:

```
inotifywait --help
# Should print usage information
```

Step 7 — Initialise Wine

REMnux includes Wine for running Windows PE binaries. Before using it you need to initialise the Wine environment, which creates the fake Windows drive structure it uses:

```
winecfg
```

A configuration window will pop up. Just click **OK** and close it. This creates the `~/.wine` directory that Wine uses as its fake `C:\` drive.

Verify it worked:

```
ls ~/.wine/drive_c/users/
# Should show a folder named after your username (remnux)
```

Step 8 — Download Firefox

REMnux does not contain a browser, so in order to access the web, we'll have to download a browser. In this case we'll be downloading Firefox

```
sudo apt install firefox -y
```

Once this completes running, you will be able to access the web through Firefox.

Step 9 — Take a snapshot

This is important. Now that the VM is set up, updated, and ready — take a clean snapshot before you do any analysis work. This lets you restore to a known-good state after each malware run.

1. In VirtualBox, go to **Machine → Take Snapshot**
2. Name it something like `clean-ready`
3. Click **OK**

To restore to this snapshot after a malware run:

1. Go to **Machine → Restore Snapshot**
2. Select `clean-ready`
3. Click **Restore**

> Get into the habit of restoring to a clean snapshot before every new sample. Never reuse a contaminated environment.

Troubleshooting

If REMnux starts with a very weird aspect ratio that looks bad to use, then click on the `View` menu in the Hypervisor bar on top, then select `Virtual Screen 1`, change the Scaling output to 100%, then change the viewport size to `1920x1080` for best experience.

If at any point during the demo you come across a tool that has not been installed for some reason, then just run `sudo apt install <package-name>`. We have taken care in order to mention all the tools that we will be using, but if there has been an oversight from our end, then follow this to get that tool up and running.

You're ready

Your REMnux VM is now set up and ready for malware analysis

REMnux official site: https://remnux.org | VirtualBox: https://www.virtualbox.org